STYLE SENSE : GENERATIVE AI–POWERED FASHION RECOMMENDATION SYSTEM

ABSTRACT

The StyleSense Generative AI-Powered Fashion Recommendation System is an advanced, intelligent web application designed to revolutionize the way users interact with fashion, personal styling, and e-commerce. In an era where personalization is key, StyleSense leverages state-of-the-art Generative AI models (such as Groq and Hugging Face) to deliver a highly tailored, interactive, and immersive styling experience.

At its core, the platform offers a multifaceted ecosystem comprising a Virtual Wardrobe, where users can digitize their clothing collection; an AI Fashion Stylist, providing on-demand, context-aware outfit recommendations based on user preferences, body type, and occasion; and an integrated Fashion Marketplace, enabling users to seamlessly discover and purchase curated items that match their unique aesthetic.

To bridge the gap between imagination and reality, StyleSense incorporates cutting-edge visual technologies, including Virtual Try-On and AI Image Generation. These features allow users to visualize themselves in recommended outfits or generate high-quality visual representations of potential new looks before making purchasing decisions. An interactive conversational AI Assistant further enhances user engagement by offering real-time fashion advice and feedback.

The system is supported by a robust architecture built on the Django framework, featuring comprehensive administrative controls, detailed analytics, user preference management, and health monitoring for third-party AI integrations. By seamlessly combining intelligent recommendation algorithms, virtual closet management, and generative media, StyleSense presents a holistic, next-generation solution for personalized fashion discovery and styling.

EXISTING SYSTEM

Traditional fashion recommendation systems primarily rely on manual browsing, rule-based filtering, or collaborative filtering techniques to suggest clothing items. Most existing fashion platforms recommend products based on a user's previous purchases, browsing history, or popular trends without considering important personal factors such as body type, skin tone, preferred fashion style, occasion, weather conditions, and seasonal preferences. These systems generally provide static recommendations and lack the capability to generate personalized styling advice or visualize complete outfits.

Many current online fashion applications do not incorporate Generative Artificial Intelligence (GenAI) to create customized outfit combinations or provide intelligent conversational assistance. They typically recommend individual products rather than complete fashion ensembles, requiring users to manually combine clothing items. Furthermore, users cannot preview AI-generated outfit designs or create virtual wardrobes to organize their preferred styles. Existing systems also provide limited personalization and fail to adapt recommendations dynamically based on changing user preferences or contextual information.

Additionally, most traditional recommendation platforms lack multimodal AI capabilities such as AI-generated fashion images, AI-generated videos, and interactive styling assistants. Shopping recommendations are generally product-centric rather than user-centric, resulting in less accurate and less engaging fashion suggestions. Administrative functionalities are often limited to basic product management and do not provide advanced analytics related to user preferences, recommendation performance, or AI-generated content.

As a result, existing systems offer a less interactive and less personalized user experience, making it difficult for users to discover outfits that truly match their individual style, body characteristics, and fashion needs.

Disadvantages of the Existing System

- Relies mainly on traditional recommendation techniques such as rule-based or collaborative filtering.
- Provides limited personalization based on browsing or purchase history only.
- Does not consider body type, skin tone, occasion, weather, or seasonal trends comprehensively.
- Lacks Generative AI capabilities for intelligent outfit creation.
- Does not provide AI-powered fashion consultation or conversational styling assistance.
- Cannot generate realistic outfit images or fashion videos.
- Offers limited visualization of recommended outfits before purchase.
- Does not support virtual wardrobe management.

- Provides static recommendations that do not adapt effectively to changing user preferences.
- Limited analytics and reporting capabilities for administrators.
- Lower user engagement due to the absence of interactive AI features.
- Requires users to manually explore and combine fashion items, increasing the time needed to select suitable outfits.

PROPOSED SYSTEM

The StyleSense: Generative AI–Powered Fashion Recommendation System is a modern web-based platform that leverages Generative Artificial Intelligence (GenAI) to provide personalized fashion recommendations, intelligent styling assistance, AI-generated outfit visualizations, and shopping suggestions. Unlike conventional recommendation systems that rely on predefined rules or collaborative filtering, the proposed system integrates Large Language Models (LLMs) through the Groq API and multimodal AI models from Hugging Face to generate context-aware, personalized, and interactive fashion experiences.

The system is developed using Python 3.11.9 and the Django Framework, with MySQL 8 serving as the backend database. It provides separate User and Admin panels with secure authentication and role-based access control. Users can create personal profiles by entering details such as body type, gender, age, height, weight, skin tone, preferred fashion style, favourite colours, and occasions. Based on these inputs, the AI recommendation engine analyzes user preferences and generates customized outfit suggestions, fashion advice, colour combinations, and seasonal styling recommendations.

The Groq API functions as the intelligent fashion assistant by processing user queries and generating natural language responses, styling tips, wardrobe suggestions, and outfit recommendations. Additionally, Hugging Face models are integrated to generate realistic AI-based fashion images and short fashion videos, enabling users to visualize recommended outfits before making purchasing decisions. The platform also includes a Virtual Wardrobe module that allows users to organize, save, and manage their favourite outfits and previously generated recommendations.

The Admin Panel enables administrators to manage users, fashion categories, occasions, seasonal trends, AI-generated content, system configurations, and application reports. It also provides analytics for monitoring AI usage, recommendation history, user engagement, and overall system performance. The integration of intelligent recommendation algorithms, multimodal AI services, and an intuitive user interface creates a scalable and efficient platform capable of delivering highly personalized fashion experiences.

The proposed system enhances the traditional fashion recommendation process by combining conversational AI, multimodal content generation, and intelligent personalization into a single platform. This approach improves user satisfaction, reduces outfit selection time, increases styling confidence, and supports informed shopping decisions while providing a modern, interactive, and responsive user experience.

Advantages of the Proposed System

- Utilizes Generative AI to provide intelligent and personalized fashion recommendations.
- Integrates the Groq API for conversational AI and styling assistance.

- Uses Hugging Face models to generate AI-powered fashion images and videos.
- Provides recommendations based on body type, skin tone, preferences, occasion, weather, and seasonal trends.
- Includes a Virtual Wardrobe for organizing and managing outfits.
- Offers personalized shopping suggestions based on generated recommendations.
- Features separate User and Admin panels with secure authentication.
- Supports responsive and modern UI/UX using Tailwind CSS and glassmorphism design.
- Stores user preferences and recommendation history securely using MySQL 8.
- Provides analytics and reporting for administrators to monitor system performance.
- Improves styling confidence and reduces the time required to select suitable outfits.
- Designed with a modular and scalable architecture for future enhancements and AI model integration.

MODULES:

1. User Authentication and Profile Management Module

This module manages user registration, login, authentication, and profile management. It securely stores user information, manages user sessions, and maintains personalized preferences required for AI-based fashion recommendations.

2. AI Fashion Recommendation Module

This module is the core of the system and analyzes user preferences, body type, favorite colors, season, occasion, and current fashion trends to generate personalized outfit recommendations and styling advice using the Groq API.

3. AI Fashion Assistant Module

This module provides a conversational AI fashion stylist that interacts with users in real time. It answers fashion-related queries, offers styling suggestions, recommends outfit combinations, and provides personalized fashion guidance based on user preferences.

4. AI Image Generation Module

This module integrates Hugging Face AI models to generate realistic fashion images based on AI-generated prompts and outfit recommendations. It enables users to visualize recommended outfits before making fashion decisions.

5. Virtual Try-On Module

This module allows users to virtually visualize AI-generated outfits on their own images. It enhances the shopping experience by providing realistic previews of clothing and accessories before purchase.

6. Virtual Wardrobe Module

This module enables users to upload, organize, and manage their personal clothing collection digitally. AI-based categorization helps users create outfit combinations using existing wardrobe items and receive personalized styling suggestions.

7. Smart Shopping Module

This module recommends fashion products, clothing items, accessories, and brands that match the user's preferences and AI-generated outfit recommendations. It also allows users to create wishlists and explore curated fashion collections.

8. AI Fashion Video Generation Module

This module utilizes Hugging Face AI models to generate short fashion showcase videos from AI-generated outfit images and styling prompts. It enables users to preview personalized fashion concepts through engaging AI-generated videos.

9. User Settings Module

This module allows users to manage their profile information, body type, fashion preferences, notification settings, privacy options, and account security. These settings are used to improve personalization and recommendation accuracy.

10. Reports and Analytics Module

This module generates detailed reports and visual analytics related to user activity, AI recommendation performance, fashion trends, API usage, and platform growth. Interactive charts help administrators monitor overall system performance.

11. Admin Portal Module

This module provides administrators with complete control over the platform. It includes user management, category management, AI monitoring, API log management, report generation, analytics, notification management, and system configuration through a secure administrative dashboard.

SYSTEM ARCHITECTURE

